REFUTATION OF THE LINEAR-SAVING CONJECTURE FOR TRIANGLE-FREE DIAMETER-FOUR COMPLETION (ERDŐS PROBLEM #619)

ARGUS OPEN RESEARCH PROJECT

Abstract. For a connected triangle-free graph G on n vertices, let $h_4(G)$ denote the minimum number of missing edges that can be added to G on the same vertex set so that the resulting supergraph $H \supseteq G$ remains triangle-free and has diameter at most 4. Problem 4.3 of Erdős, Gyárfás, and Ruszinkó (1998; Erdős Problem #619) asks whether there exists a constant $\epsilon > 0$ such that $h_4(G) \leq (1 - \epsilon)n$ for every connected triangle-free graph G on n vertices. We document and verify the Lean 4 disproof showing that for every $\eta \in (0, 1)$ and all sufficiently large n, there exist connected triangle-free graphs G on n vertices with $h_4(G) \geq (1 - \eta)n$.

1. Main Theorem

Theorem 1.1 (Formally Verified in Lean 4)**.** *For every real $\eta \in (0, 1)$, there exists $N_0(\eta)$ such that for every $n \geq N_0(\eta)$ there is a connected triangle-free graph G on n vertices (constructed by attaching pendant vertices to a dense triangle-free core of independence number $o(n)$) such that every triangle-free supergraph $H \supseteq G$ on $V(G)$ with* $\operatorname{diam}(H) \leq 4$ *satisfies*

$$|E(H) \setminus E(G)| \;\geq\; (1 - \eta)n,$$

and therefore $h_4(G) \geq (1 - \eta)n$, refuting the existence of a uniform constant $\epsilon > 0$ with $h_4(G) \leq (1 - \epsilon)n$.

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